

mongo_db_us_housing

June 6, 2026

```
[40]: from pymongo import MongoClient
import pandas as pd
```

```
[41]: # connect to MongoDB
client = MongoClient('mongodb://localhost:27017')
```

```
[42]: db = client['us_housing_db']
```

```
[43]: collection = db['us_housing']
```

```
[44]: records = collection.find().limit(4)
for record in records:
    print(record, '\n')
```

```
{'_id': ObjectId('6a249dc0621633790618348b'), 'Avg': {'Area Income':
79545.45857431678, 'Area House Age': 5.682861321615587, 'Area Number of
Rooms': 7.009188142792237, 'Area Number of Bedrooms': 4.09}, 'Area Population':
23086.800502686456, 'Price': 1059033.5578701235, 'Address': '208 Michael Ferry
Apt. 674\nLaurabury, NE 37010-5101'}
```

```
{'_id': ObjectId('6a249dc0621633790618348c'), 'Avg': {'Area Income':
79248.64245482568, 'Area House Age': 6.0028998082752425, 'Area Number of
Rooms': 6.730821019094919, 'Area Number of Bedrooms': 3.09}, 'Area Population':
40173.07217364482, 'Price': 1505890.91484695, 'Address': '188 Johnson Views
Suite 079\nLake Kathleen, CA 48958'}
```

```
{'_id': ObjectId('6a249dc0621633790618348d'), 'Avg': {'Area Income':
61287.067178656784, 'Area House Age': 5.865889840310001, 'Area Number of
Rooms': 8.512727430375099, 'Area Number of Bedrooms': 5.13}, 'Area Population':
36882.15939970458, 'Price': 1058987.9878760849, 'Address': '9127 Elizabeth
Stravenue\nDanieltown, WI 06482-3489'}
```

```
{'_id': ObjectId('6a249dc0621633790618348e'), 'Avg': {'Area Income':
63345.24004622798, 'Area House Age': 7.1882360945186425, 'Area Number of
Rooms': 5.586728664827653, 'Area Number of Bedrooms': 3.26}, 'Area Population':
34310.24283090706, 'Price': 1260616.8066294468, 'Address': 'USS Barnett\nFPO AP
44820'}
```

```
[45]: # find a property based on address
result = collection.find_one({"Address": '208 Michael Ferry Apt. 674\nLaurabury, NE 37010-5101'})
print(result)
```

```
{'_id': ObjectId('6a249dc0621633790618348b'), 'Avg': {'Area Income': 79545.45857431678, 'Area House Age': 5.682861321615587, 'Area Number of Rooms': 7.009188142792237, 'Area Number of Bedrooms': 4.09}, 'Area Population': 23086.800502686456, 'Price': 1059033.5578701235, 'Address': '208 Michael Ferry Apt. 674\nLaurabury, NE 37010-5101'}
```

```
[69]: # find property based on price less than $840,000
query = {"Price": {"$gt": 840000}}

results = collection.find(query)

df = pd.DataFrame(list(results))
print(df.head())
print(df.shape)
```

```
      _id \
0  6a249dc0621633790618348b
1  6a249dc0621633790618348c
2  6a249dc0621633790618348d
3  6a249dc0621633790618348e
4  6a249dc06216337906183490
```

```
      Avg Area Population \
0  {'Area Income': 79545.45857431678, 'Area Hou... 23086.800503
1  {'Area Income': 79248.64245482568, 'Area Hou... 40173.072174
2  {'Area Income': 61287.067178656784, 'Area Ho... 36882.159400
3  {'Area Income': 63345.24004622798, 'Area Hou... 34310.242831
4  {'Area Income': 80175.7541594853, 'Area Hous... 26748.428425
```

```
      Price Address
0  1.059034e+06 208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1  1.505891e+06 188 Johnson Views Suite 079\nLake Kathleen, CA...
2  1.058988e+06 9127 Elizabeth Stravenue\nDanielstown, WI 06482...
3  1.260617e+06 USS Barnett\nFPO AP 44820
4  1.068138e+06 06039 Jennifer Islands Apt. 443\nTracyport, KS...
(4328, 5)
```

```
[68]: # find property based on area population less than 3,000
query = {"Area Population": {"$gt": 3000}}

results = collection.find(query)
```

```
df = pd.DataFrame(list(results))
print(df.head)
```

```
<bound method NDFrame.head of                                _id \
0      6a249dc0621633790618348b
1      6a249dc0621633790618348c
2      6a249dc0621633790618348d
3      6a249dc0621633790618348e
4      6a249dc0621633790618348f
...
4994  6a249dc0621633790618480e
4995  6a249dc0621633790618480f
4996  6a249dc06216337906184810
4997  6a249dc06216337906184811
4998  6a249dc06216337906184812

                                Avg  Area Population \
0      {' Area Income': 79545.45857431678, ' Area Hou...  23086.800503
1      {' Area Income': 79248.64245482568, ' Area Hou...  40173.072174
2      {' Area Income': 61287.067178656784, ' Area Ho...  36882.159400
3      {' Area Income': 63345.24004622798, ' Area Hou...  34310.242831
4      {' Area Income': 59982.197225708034, ' Area Ho...  26354.109472
...
4994  {' Area Income': 60567.944139865984, ' Area Ho...  22837.361035
4995  {' Area Income': 78491.27543459789, ' Area Hou...  25616.115489
4996  {' Area Income': 63390.68688550463, ' Area Hou...  33266.145490
4997  {' Area Income': 68001.33123508695, ' Area Hou...  42625.620156
4998  {' Area Income': 65510.581803666224, ' Area Ho...  46501.283803

                                Price                                Address
0      1.059034e+06  208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1      1.505891e+06  188 Johnson Views Suite 079\nLake Kathleen, CA...
2      1.058988e+06  9127 Elizabeth Stravenue\nDanielstown, WI 06482...
3      1.260617e+06                                USS Barnett\nFPO AP 44820
4      6.309435e+05                                USNS Raymond\nFPO AE 09386
...
4994  1.060194e+06                                USNS Williams\nFPO AP 30153-7653
4995  1.482618e+06                                PSC 9258, Box 8489\nAPO AA 42991-3352
4996  1.030730e+06  4215 Tracy Garden Suite 076\nJoshualand, VA 01...
4997  1.198657e+06                                USS Wallace\nFPO AE 73316
4998  1.298950e+06  37778 George Ridges Apt. 509\nEast Holly, NV 2...
```

```
[4999 rows x 5 columns]>
```

```
[72]: df.columns
```

```
[72]: Index(['_id', 'Avg', 'Area Population', 'Price', 'Address'], dtype='object')
```

```
[73]: print(f'Total number of records: {collection.count_documents({}})')
```

Total number of records: 5000

```
[84]: most_expensive = collection.find_one(sort=[('Price',-1)])
      for index, value in most_expensive.items():
          print(index, value)
```

_id 6a249dc0621633790618365c
Avg {' Area Income': 90592.46960928394, ' Area House Age': 7.700132109848975, ' Area Number of Rooms': 9.708803014905254, ' Area Number of Bedrooms': 5.19}
Area Population 37223.87616726714
Price 2469065.5941747027
Address USNS Vargas
FPO AE 56319-6904

```
[93]: pipeline = [
      {
          '$group':{
              '_id': None,
              'average_price': {'$avg': '$Price'}
          }
      ]

      result = list(collection.aggregate(pipeline))

      average_price = result[0]['average_price']
      print(f'Average Price: {average_price}')
```

Average Price: 1232072.654142357

```
[ ]:
```